

CODEX CHAT

Create a python script for this 6m x 6m 3D cube using this coordinates of nodes. The z and x axis as the lateral and y as the vertical axis. Node 1 is at coordinates 0,0,0. Make it navigatable.

Tuesday 11:17 PM
Worked for 43s

Created [cube_nodes_6m.py](C:\\Users\\stewa\\python-practice\\lab01\\cube_nodes_6m.py).

It defines all 8 nodes for the 6 m cube, the 12 frame members shown, and opens an interactive Matplotlib 3D view. Run:

```
python cube_nodes_6m.py
```

Use drag to rotate, mouse wheel to zoom, and right-drag to pan.

Edited cube_nodes_6m.py+77-0Review changes
UndoReview
Tuesday 11:18 PM

keep in mind that z and x axis are the lateral and y axis is vertical

Tuesday 11:20 PM

Understood. The script already follows that convention:

- x: lateral
- z: lateral
- y: vertical (height)

So node 1 is $(0, 0, 0)$, and the roof nodes use $y = 6$.

Tuesday 11:20 PM

the placement of "x lateral (m)" is in the wrong axis

Tuesday 11:22 PM
Worked for 30s

Fixed the displayed axis placement in [cube_nodes_6m.py](C:\\Users\\stewa\\python-practice\\lab01\\cube_nodes_6m.py).

Y vertical (m) is now the upright plotted axis; X lateral (m) and Z lateral (m) form the horizontal ground plane. Node coordinates remain (x, y, z) .

Edited cube_nodes_6m.py+9-7Review changes
UndoReview
Tuesday 11:23 PM

Improve the existing Python solver by adding the following features, similar to RISA or STAAD:

1. Add support conditions at the bottom nodes. For now, use pinned supports.
2. Add a beta angle.

3. Add local axes and global

4. Add degrees of freedom (DOF) for each node.

5. Add degrees of freedom for beam pinned at x direction moment at z direction

Show Pinned symbol if beam is pinned.

5. Update the structural diagram.

6. Update the Excel output file.

7. Save the updated Python solver as **Rev. 1** and check the syntax and grammar.

Show more
Tuesday 11:30 PM
Worked for 2m 32s

Created Rev. 1: [cube_nodes_6m_Rev_1.py](C:\\Users\\stewa\\python-practice\\lab01\\cube_nodes_6m_Rev_1.py), plus generated Excel workbook: [structural_model_Rev_1.xlsx](C:\\Users\\stewa\\python-practice\\lab01\\structural_model_Rev_1.xlsx).

Included:

- Pinned supports at nodes 1–4: U_x , U_y , U_z restrained; rotations free.
- Six global node DOFs: U_x , U_y , U_z , R_x , R_y , R_z .
- Member beta-angle field (β_{deg}).
- Calculated local member axes and displayed global/local axis triads.
- Pinned beam ends releasing global R_z (moment about Z), shown as magenta hollow circles.
- Updated interactive 3D structural diagram with pinned-support symbols.
- Excel sheets for Nodes, Members, and Node DOFs.

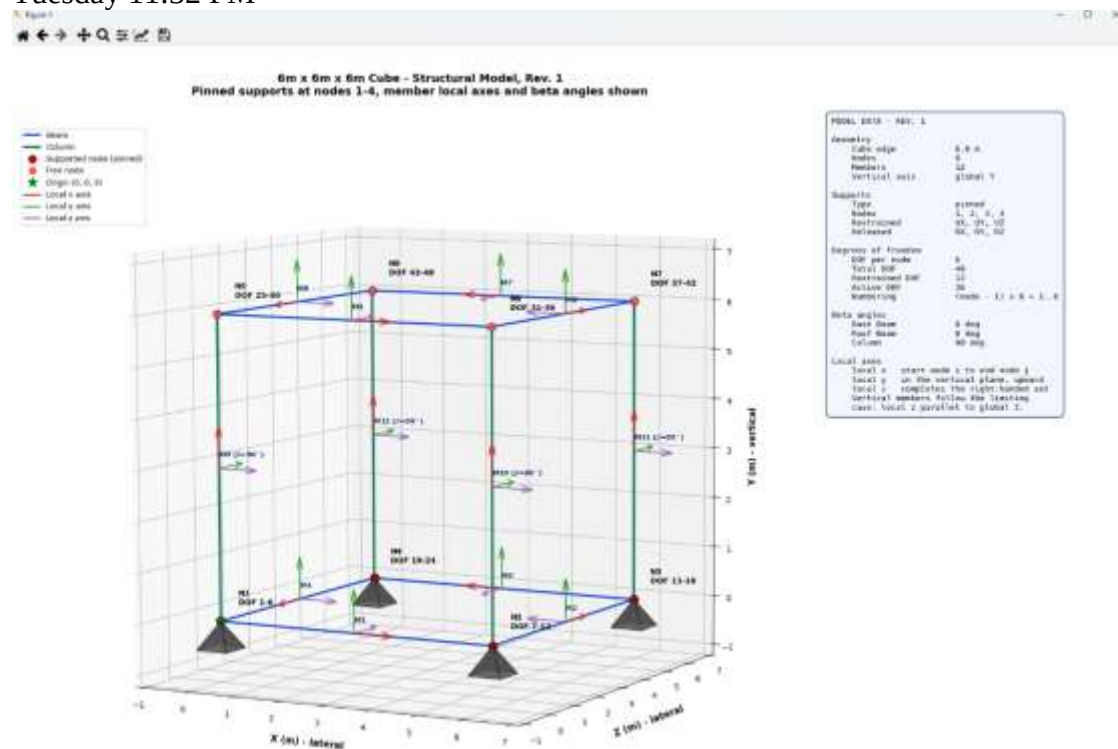
Validated syntax, generated workbook integrity, and XML parsing successfully. Run with:

python cube_nodes_6m_Rev_1.py

Edited cube_nodes_6m_Rev_1.py+243-0Review changes

UndoReview

Tuesday 11:32 PM



Make the output appear like this

Tuesday 11:37 PM

I'll reshape the Rev. 1 plot around the reference layout: blue beams, green columns, support pyramids, node/DOF labels, red-green-purple local-axis arrows, and a right-side model-data panel. I'll also set the column beta angle to 90° as shown in your target.

Edited a file

Edited cube_nodes_6m_Rev_1.py+96-51Review changes

UndoReview

	A	B	C	D	E	F	G	H	I	J	K	L
1	Node	Local DOF	DOF	Description	Global DOF No.	Status	Equation No.					
2	2	5	RY	Rotation Y	11	Active	5					
3	2	6	RZ	Rotation Z	12	Active	6					
4	3	1	UX	Translation X	13	Restrained	-					
5	3	2	UY	Translation Y	14	Restrained	-					
6	3	3	UZ	Translation Z	15	Restrained	-					
7	3	4	RX	Rotation X	16	Active	7					
8	3	5	RY	Rotation Y	17	Active	8					
9	3	6	RZ	Rotation Z	18	Active	9					
10	4	1	UX	Translation X	19	Restrained	-					
11	4	2	UY	Translation Y	20	Restrained	-					
12	4	3	UZ	Translation Z	21	Restrained	-					
13	4	4	RX	Rotation X	22	Active	10					
14	4	5	RY	Rotation Y	23	Active	11					
15	4	6	RZ	Rotation Z	24	Active	12					
16	5	1	UX	Translation X	25	Active	13					
17	5	2	UY	Translation Y	26	Active	14					
18	5	3	UZ	Translation Z	27	Active	15					
19	5	4	RX	Rotation X	28	Active	16					
20	5	5	RY	Rotation Y	29	Active	17					
21	5	6	RZ	Rotation Z	30	Active	18					
22	6	1	UX	Translation X	31	Active	19					
23	6	2	UY	Translation Y	32	Active	20					
24	6	3	UZ	Translation Z	33	Active	21					
25	6	4	RX	Rotation X	34	Active	22					
26	6	5	RY	Rotation Y	35	Active	23					
27	6	6	RZ	Rotation Z	36	Active	24					
28	7	1	UX	Translation X	37	Active	25					
29	7	2	UY	Translation Y	38	Active	26					
30	7	3	UZ	Translation Z	39	Active	27					
31	7	4	RX	Rotation X	40	Active	28					
32	7	5	RY	Rotation Y	41	Active	29					
33	7	6	RZ	Rotation Z	42	Active	30					
34	8	1	UX	Translation X	43	Active	31					
35	8	2	UY	Translation Y	44	Active	32					
36	8	3	UZ	Translation Z	45	Active	33					
37	8	4	RX	Rotation X	46	Active	34					
38	8	5	RY	Rotation Y	47	Active	35					
39	8	6	RZ	Rotation Z	48	Active	36					
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<	>	Model Summary	Nodes	Member Incidences	Supports	Local Axes	Node DOF	DOF Numbering
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	A	B	C	D	E	F	G	H	I	J	K	L
1	Node	Support	Global DOF Range	UX	UY	UZ	RX	RY	RZ	Active DOF		
2	1	Pinned	1 - 6	Restrained	Restrained	Restrained	Active	Active	Active	3		
3	2	Pinned	7 - 12	Restrained	Restrained	Restrained	Active	Active	Active	3		
4	3	Pinned	13 - 18	Restrained	Restrained	Restrained	Active	Active	Active	3		
5	4	Pinned	19 - 24	Restrained	Restrained	Restrained	Active	Active	Active	3		
6	5	Free	25 - 30	Active	Active	Active	Active	Active	Active	6		
7	6	Free	31 - 36	Active	Active	Active	Active	Active	Active	6		
8	7	Free	37 - 42	Active	Active	Active	Active	Active	Active	6		
9	8	Free	43 - 48	Active	Active	Active	Active	Active	Active	6		
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	A	B	C	D	E	F	G	H	I	J
1	Member	Node i (Start)	Node j (End)	Type	Length (m)	Beta (deg)				
2	1	1	2	Base Beam	6.000	0.0				
3	2	2	3	Base Beam	6.000	0.0				
4	3	3	4	Base Beam	6.000	0.0				
5	4	4	1	Base Beam	6.000	0.0				
6	5	5	6	Roof Beam	6.000	0.0				
7	6	6	7	Roof Beam	6.000	0.0				
8	7	7	8	Roof Beam	6.000	0.0				
9	8	8	5	Roof Beam	6.000	0.0				
10	9	1	5	Column	6.000	90.0				
11	10	2	6	Column	6.000	90.0				
12	11	3	7	Column	6.000	90.0				
13	12	4	8	Column	6.000	90.0				
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Model Summary
Nodes
Member Incidences
Supports
Local Axes
Node DOF
DOF Numbering

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Node	X (m)	Y (m)	Z (m)	Support								
2	1	0.000	0.000	0.000	Pinned								
3	2	6.000	0.000	0.000	Pinned								
4	3	6.000	0.000	6.000	Pinned								
5	4	0.000	0.000	6.000	Pinned								
6	5	0.000	6.000	0.000	Free								
7	6	6.000	6.000	0.000	Free								
8	7	6.000	6.000	6.000	Free								
9	8	0.000	6.000	6.000	Free								
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Model Summary

Nodes

Member Incidences

Supports

Local Axes

Node DOF

DOF Numbering

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Ready

Accessibility: Good to go

	A	B	C	D	E	F	G	H	I	J
1	Item	Value								
2	Revision	Rev. 1								
3	Cube edge length (m)	6								
4	Number of nodes	8								
5	Number of members	12								
6	Supported nodes	4								
7	Support type	Pinned								
8	DOF per node	6								
9	Total DOF	48								
10	Restrained DOF	12								
11	Active DOF (equations)	36								
12	Global vertical axis	Y								
13	Global lateral axes	X and Z								
14	Beta angle, base beams (deg)	0								
15	Beta angle, roof beams (deg)	0								
16	Beta angle, columns (deg)	90								
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I also need to see this kind of datas in the excel files

Tuesday 11:42 PM
Worked for 3m 27s

Updated Rev. 1 to produce the requested RISA/STAAD-style diagram and detailed Excel data.

The new workbook is [structural_model_Rev_1_detailed.xlsx](C:\\Users\\stewa\\python-practice\\lab01\\structural_model_Rev_1_detailed.xlsx), with these sheets:

- Model Summary
- Nodes
- Member Incidences
- Supports
- Local Axes
- Node DOF
- DOF Numbering

It includes pinned/free statuses, restraint codes, all 48 global DOFs, 36 active equation numbers, member lengths, beta angles, and local-axis components. Headers and Active/Restrained cells are formatted like the examples.

The existing `structural_model_Rev_1.xlsx` was open and locked by Excel, so the expanded output was saved separately. Running `[cube_nodes_6m_Rev_1.py](C:\\Users\\stewa\\python-practice\\lab01\\cube_nodes_6m_Rev_1.py)` regenerates the detailed workbook.

I checked this by analyzing the output data to confirm that it shows relevant values and follows the format given by the instructor.